

Jinay Patel

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EDUCATION

University of Waterloo

Bachelor of Computer Engineering

Expected Graduation: Jun 2031

SKILLS

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years)

Technologies: PyTorch, Langchain, Mastra, CMake, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity

EXPERIENCE

Lead Software Engineer

October 2023 — June 2026

Seaquam Robotics

Delta, BC

- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by +50%
- Created a custom React and Tauri CMS interface, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server response speed by +150%

Embedded Controls Software Engineer

October 2021 — June 2026

Seaquam Robotics

Delta, BC

- Implemented C++ motion algorithms, securing a 100% win rate during autonomous periods across 15+ matches
- Designed an Interpolated Look Up Table using a Monotonic Cubic Spline for tuning PID constants in real-time which resulted in a 200% increase in accuracy across all movements with error tolerances being <1 inch

Software Engineering Tutor

September 2025 — March 2026

Code Ninjas

Surrey, BC

- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving our local branch search visibility

Freelance Systems Administrator / IT Consultant

Convergence Media

Remote

- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a Tailscale tunnel to expose a public link and allow anyone to access the Nextcloud server for video calls or file sharing

PROJECTS

Telemetry Data Visualization

- Takes in telemetry data on the robot's position and error in a system through a **websocket** and then visualizes it in **real-time on a Svelte dashboard**.
- Uses **Chart.js** for displaying the graphs in real-time, and the **Canvas API** for drawing the robot on a 2D field.

AI Complaint Categorization System

- Analyzes text, audio, and image based complaints and then categorizes each of them. It all get stored in a **vector database** for a **RAG pipeline**. This project was a **hackathon finalist at the Headerstarter Fellowship Hackathon**.
- Makes use of **OCR Space**, **Gemini**, **Hugging Face**, **Langchain**, and **Supabase** (for the vector database).

Content Management System & Website Backend

- Created a backend to **aggregate data from an API** and a custom **content management system** which allowed our coach to update data on our robotics website.
- The backend was written using **Deno** and the content management system was created with **Tauri** and **React**.

AWARDS

Excellence Award @ Vex Robotics World Championship, RECF & Vex Robotics

April 2026

Awarded to the top team out of 20,000+ teams worldwide for excellent performance

Excellence Award @ Provincial Championship, Vex Robotics

February 2026

Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks

July 2025

Awarded for developing [Posturai](#), a real-time posture correction software

